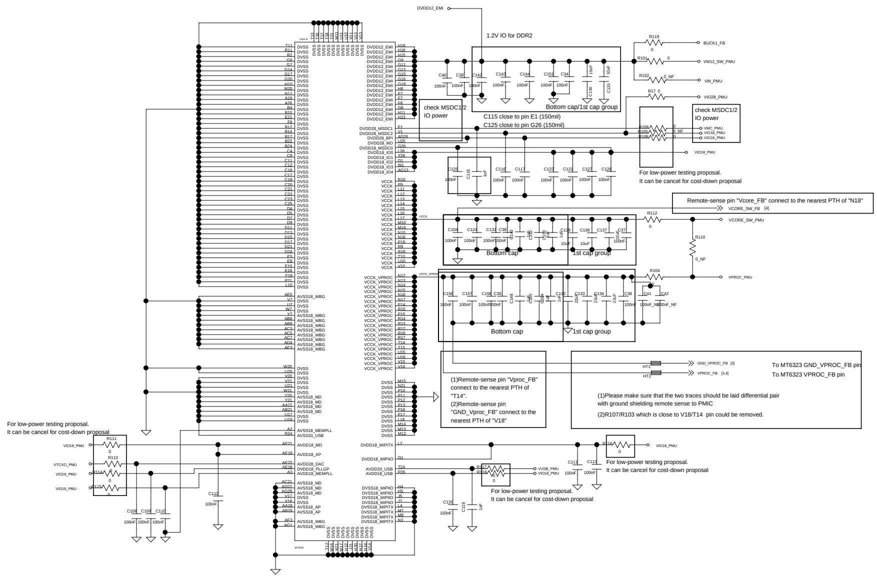
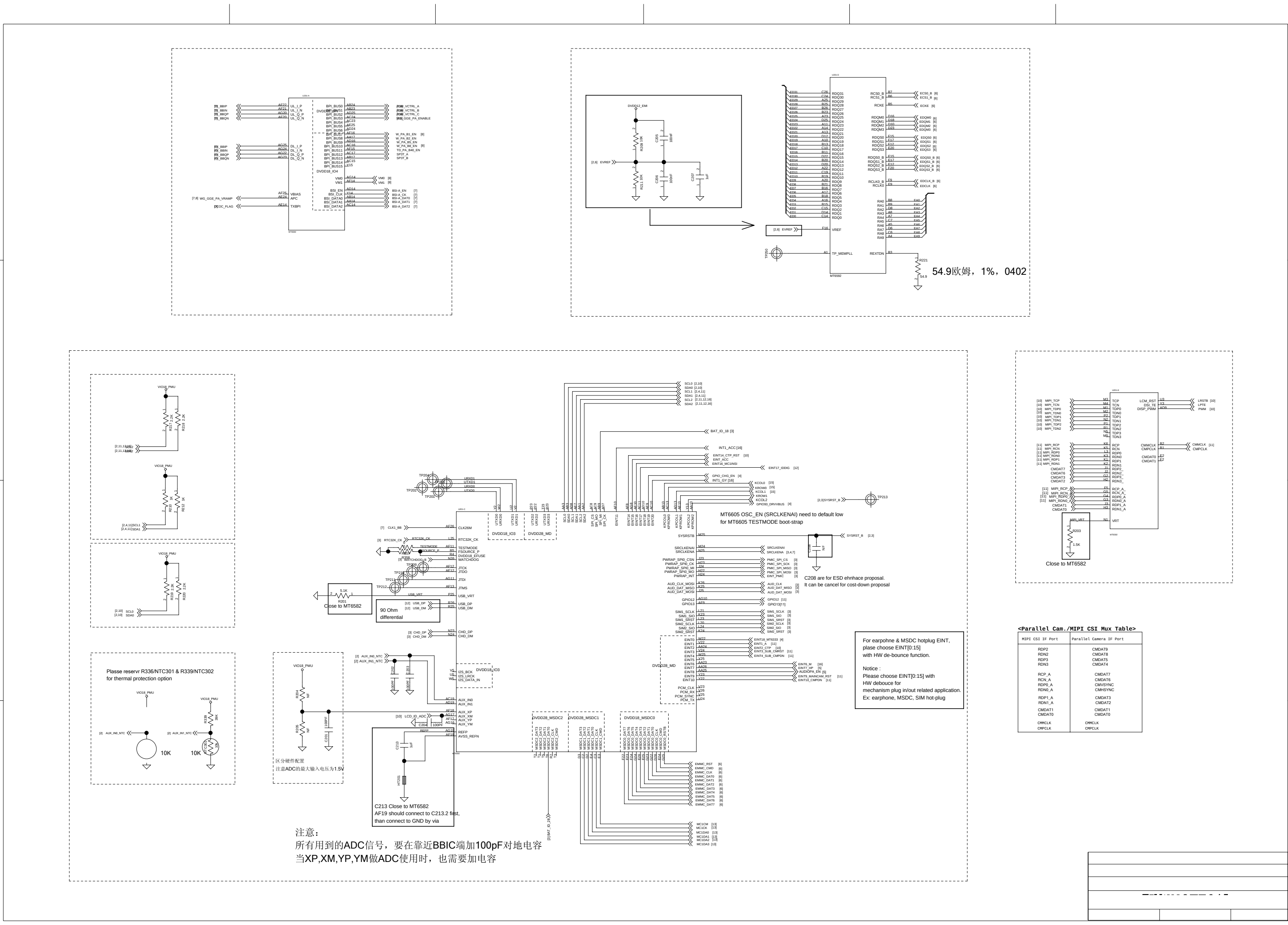


MT6582 R119=NF, R101=NF, R102=0, R112=0, R110=0, R104=0

MT6592 R119=0, R101=0, R102=NF, R112=0, R110=NF, R104=0

MT6582 C135, C137, 可以NF

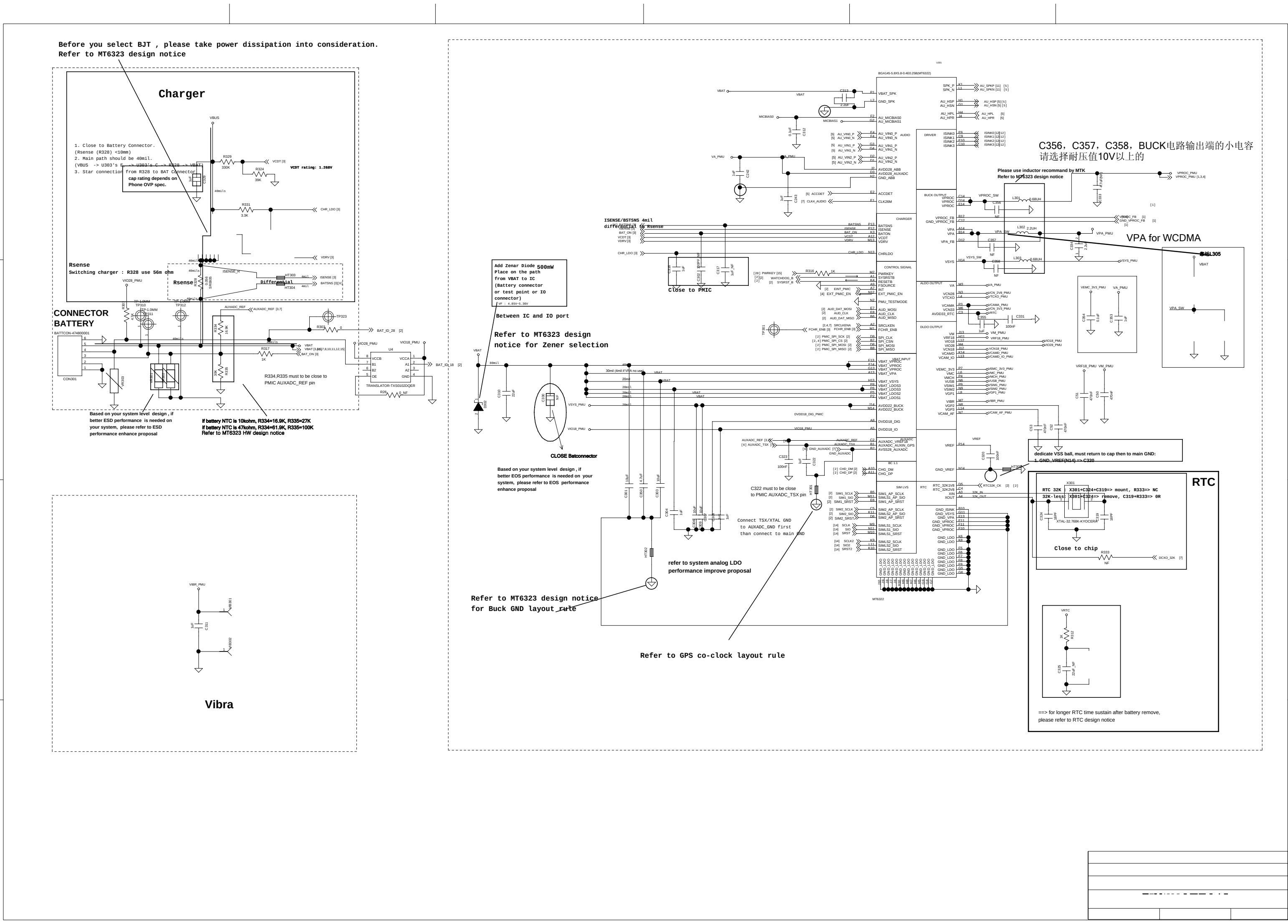
MT6592 C135=22uF, C137=22uF

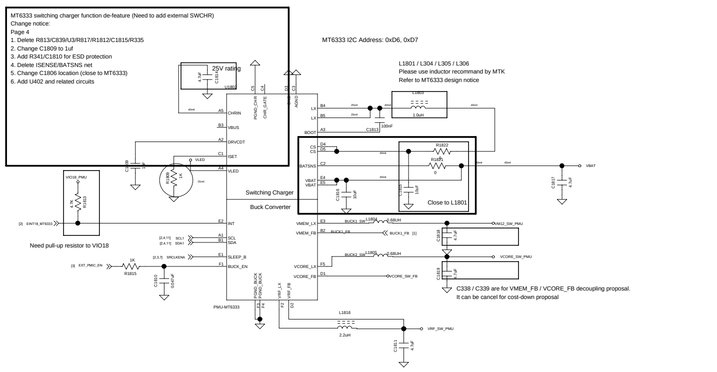


<Parallel Cam./MIPI CSI Mux Table>

MIPI CSI IF Port	Parallel Camera IF Port
RDP2	CMDAT9
RDN2	CMDAT8
RDP3	CMDAT5
RDN3	CMDAT4
RCP_A	CMDAT7
RCN_A	CMDAT6
RDP0_A	CMVSYNC
RDN0_A	CMHSYNC
RDP1_A	CMDAT3
RDN1_A	CMDAT2
CMDAT1	CMDAT1
CMDAT0	CMDAT0
CMMCLK	CMMCLK
CMPCLK	CMPCLK

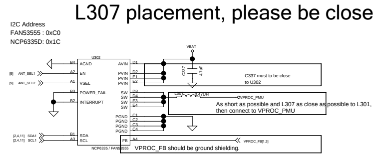
Regulator	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VPROC	0.7~1.4	2800	>10uF	L=0.68uH,C=10uF*4	Total outptu cap>40uF
VSYS	2.2	1200	>10uF	L=0.68uH,C=10uF*2	Total outptu cap>20uF
VPA	0.5~3.4	600	>4.7uF	L=2.2uH,C=2.2uF+2.2uF	Output cap range 4.4uF +/-20%
LDO	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VM	1.24 /1.39/1.54/1.84	700	10uF	-20%~+20%	Far-end bypass cap
VRF18	1.825	200	1uF	-20%~+200%	Far-end bypass cap
VIO18	1.8	300	4.7uF	-20%~+200%	Far-end bypass cap
VCN18	1.8	120	1uF	-20%~+20%	Far-end bypass cap
VCAMD	1.2 /1.3/1.5/1.8	150	1uF	-20%~+20%	Far-end bypass cap
VCAM_IO	1.8	100	1uF	-20%~+20%	Far-end bypass cap
VGP3	1.2 /1.3/1.5/1.8	200	1uF	-20%~+20%	Far-end bypass cap
VA	2.8	150	1uF	-20%~+20%	Far-end bypass cap
VTCXO	2.8	40	1uF	-20%~+20%	Far-end bypass cap
VCN28	2.8	30	1uF	-20%~+20%	Far-end bypass cap
VCAMA	2.8	150	3.2uF	-20%~+20%	1uF near-end ,2.2uF Far-end bypass cap
VCN33	3.3/3.4/3.5/3.6	240(MT6323) 350(MT6322)	4.7uF	-20%~+20%	Far-end bypass cap
VIO28	2.8	200	2.2uF	-20%~+200%	Far-end bypass cap
VUSB	3.3	20	1uF	-20%~+20%	Far-end bypass cap
VMC	1.8 /3.3	100	1uF	-20%~+20%	Far-end bypass cap
VMCH	3.0 /3.3	400	2.2uF	-20%~+20%	Far-end bypass cap
VEMC_3V3	3.0 /3.3	400	4.7uF	-20%~+20%	Far-end bypass cap
VCAM_AF	1.2/1.3/1.5/1.8 /2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VSIM1	1.8 /3.0	50	1uF	-20%~+20%	Far-end bypass cap
VSIM2	1.8 /3.0	50	1uF	-20%~+20%	Far-end bypass cap
VGP1	1.2/1.3/1.5/1.8/2.0 2.8 /3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VGP2	1.2/1.3/1.5/1.8/2.0 2.5 /2.8/3.0	100	1uF	-20%~+20%	Far-end bypass cap
VIBR	1.2 /1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VDIG18	1.8	20	1uF	-20%~+20%	Far-end bypass cap
VRTC	2.8	2	0.1uF to 1000uF	-20%~+20%	Far-end bypass cap





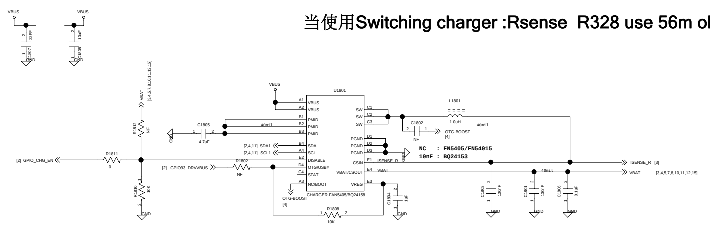
PMU-MT6333

U302 placement, please be close to MT6322
L307 placement, please be close to L301



External DC-DC for VPROC

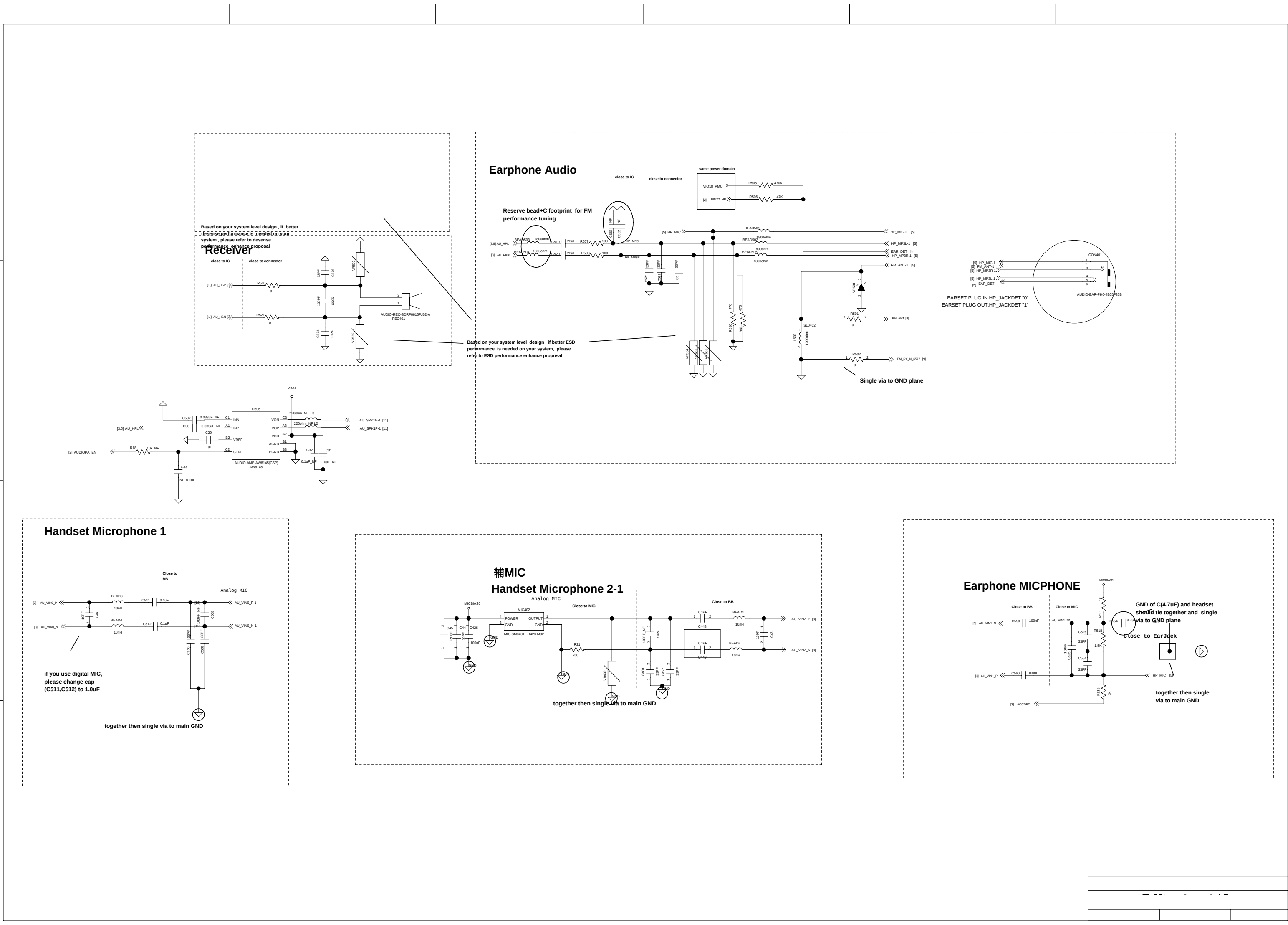
当使用Switching charger :Rsense R328 use 56m ohm



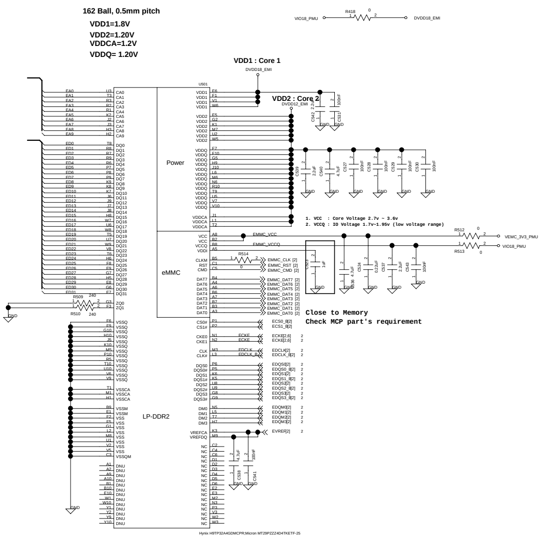
FN5405   R1810=10K,R1812=NF, C1802=NF
FN5402   R1810=NF, R1812=10K,C1802=NF
BQ24158   R1810=10K,R1812=NF, C1802=10nF

If switching charger is used:
(1) R1801~R1805, C1801~C1806, L1801, U1801 are needed
(2) U303, U304 change to NC
(3) R328 change to 56m Ohm

Switching Charger

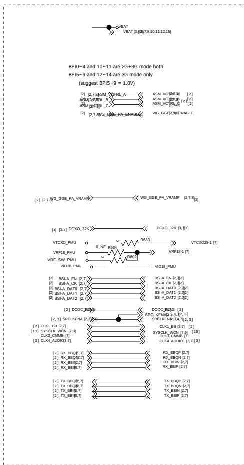

Earphone Audio
close to IC
close to connector
same power domain
Reserve bead+C footprint for FM performance tuning
Based on your system level design , if better desense performance is needed on your system , please refer to desense performance enhance proposal
Receiver
Based on your system level design , if better ESD performance is needed on your system, please refer to ESD performance enhance proposal
Single via to GND plane
EARSET PLUG IN:HP_JACKDET "0"
EARSET PLUG OUT:HP_JACKDET "1"
Handset Microphone 1
Close to BB
Analog MIC
if you use digital MIC, please change cap (C511,C512) to 1.0uF
together then single via to main GND
辅MIC
Handset Microphone 2-1
Analog MIC
Close to MIC
Close to BB
together then single via to main GND
Earphone MICPHONE
Close to BB
Close to MIC
GND of C(4.7uF) and headset should tie together and single via to GND plane
Close to EarJack
together then single via to main GND

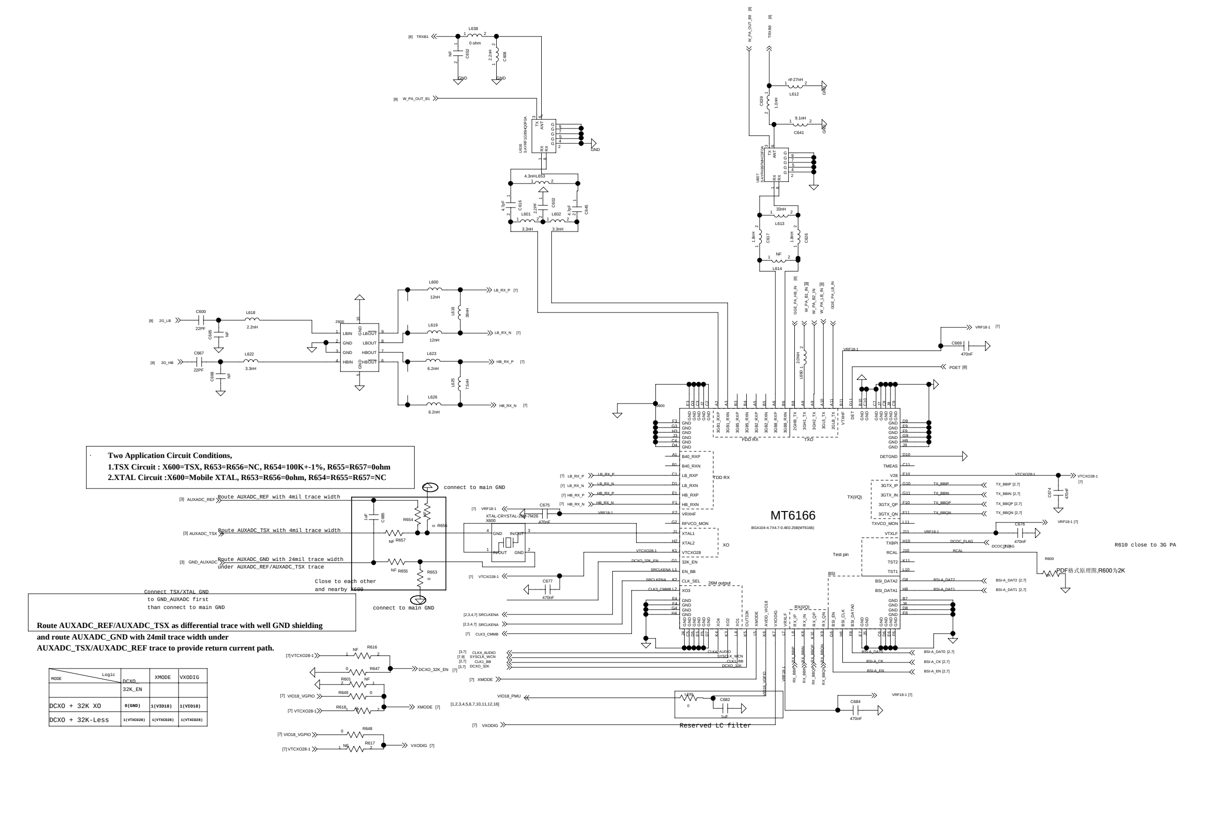
eMMC+LPDDR2


162 Ball, 0.5mm pitch
VDD1=1.8V
VDD2=1.20V
VDDCA=1.2V
VDDQ= 1.20V
VIO18_PMU
R418
0
DVDD18_EMI
VDD1 : Core 1
VDD2 : Core 2
DVDD12_EMI
U501
Power
eMMC
LP-DDR2
1. VCC : Core Voltage 2.7v ~ 3.6v
2. VCCQ : IO Voltage 1.7v~1.95v (low voltage range)
EMMC_VCC
EMMC_VCCQ
R512
R513
VEMC_3V3_PMU
EMMC_CLK [2]
EMMC_RST [2]
EMMC_CMD [2]
EMMC_DAT7 [2]
EMMC_DAT6 [2]
EMMC_DAT5 [2]
EMMC_DAT4 [2]
EMMC_DAT3 [2]
EMMC_DAT2 [2]
EMMC_DAT1 [2]
EMMC_DAT0 [2]
Close to Memory
Check MCP part's requirement

RF9810 control logic table

	Enable	VctC	VctB	VctA
LB_GMSK_TX	H	L	H	L
HB_GMSK_TX	H	L	H	H
LB_EDGE_TX	H	L	L	L
HB_EDGE_TX	H	L	L	H
RX1	L	H	L	L
RX2	L	L	H	L
RX3	L	L	H	H
RX4	L	L	L	H





Two Application Circuit Conditions,

1.TSX Circuit : X600=TSX, R653=R656=NC, R654=100K+-1%, R655=R657=0ohm

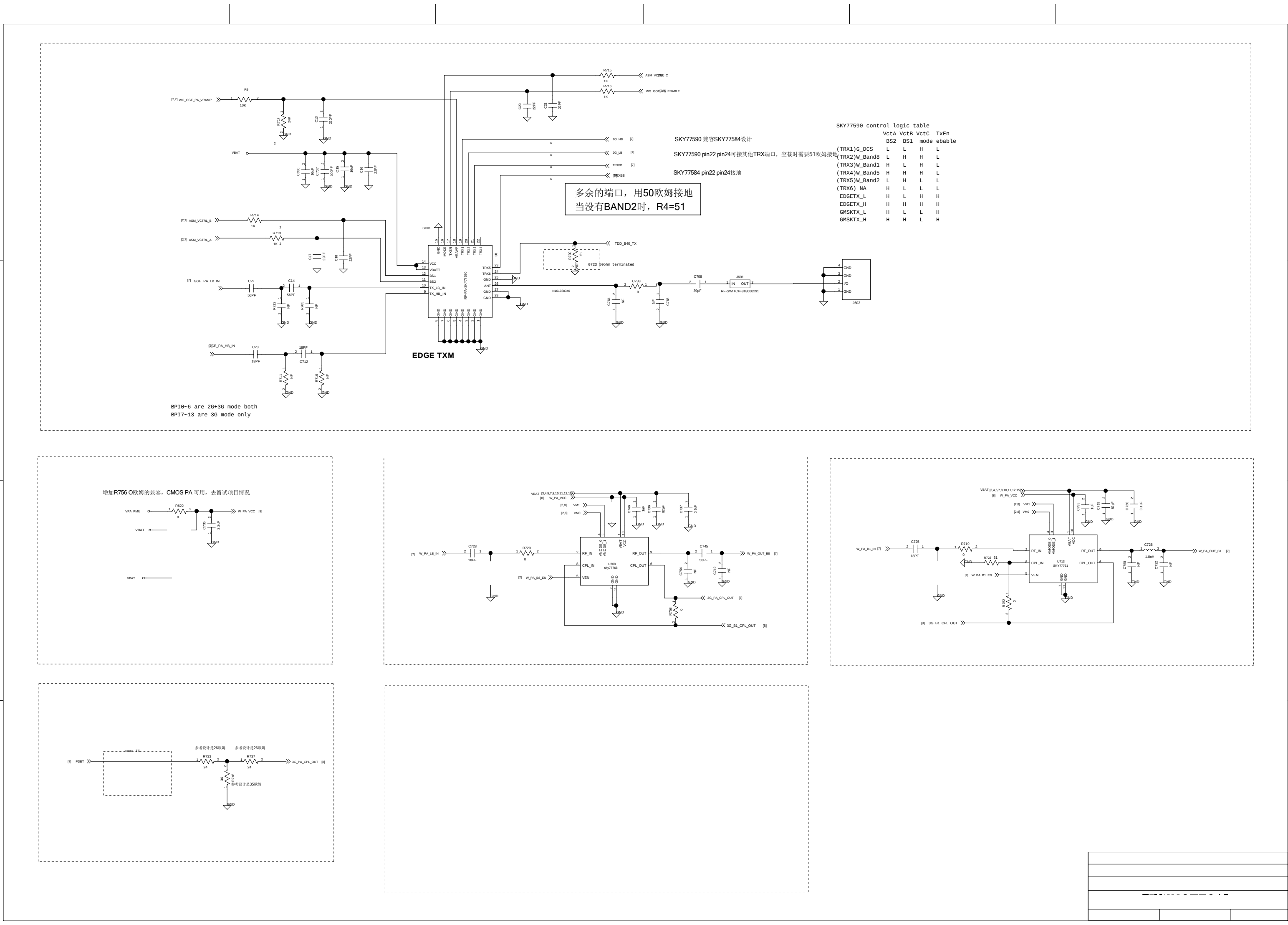
2.XTAL Circuit :X600=Mobile XTAL, R653=R656=0ohm, R654=R655=R657=NC

Connect TSX/XTAL GND
to GND_AUXADC first
than connect to main GND

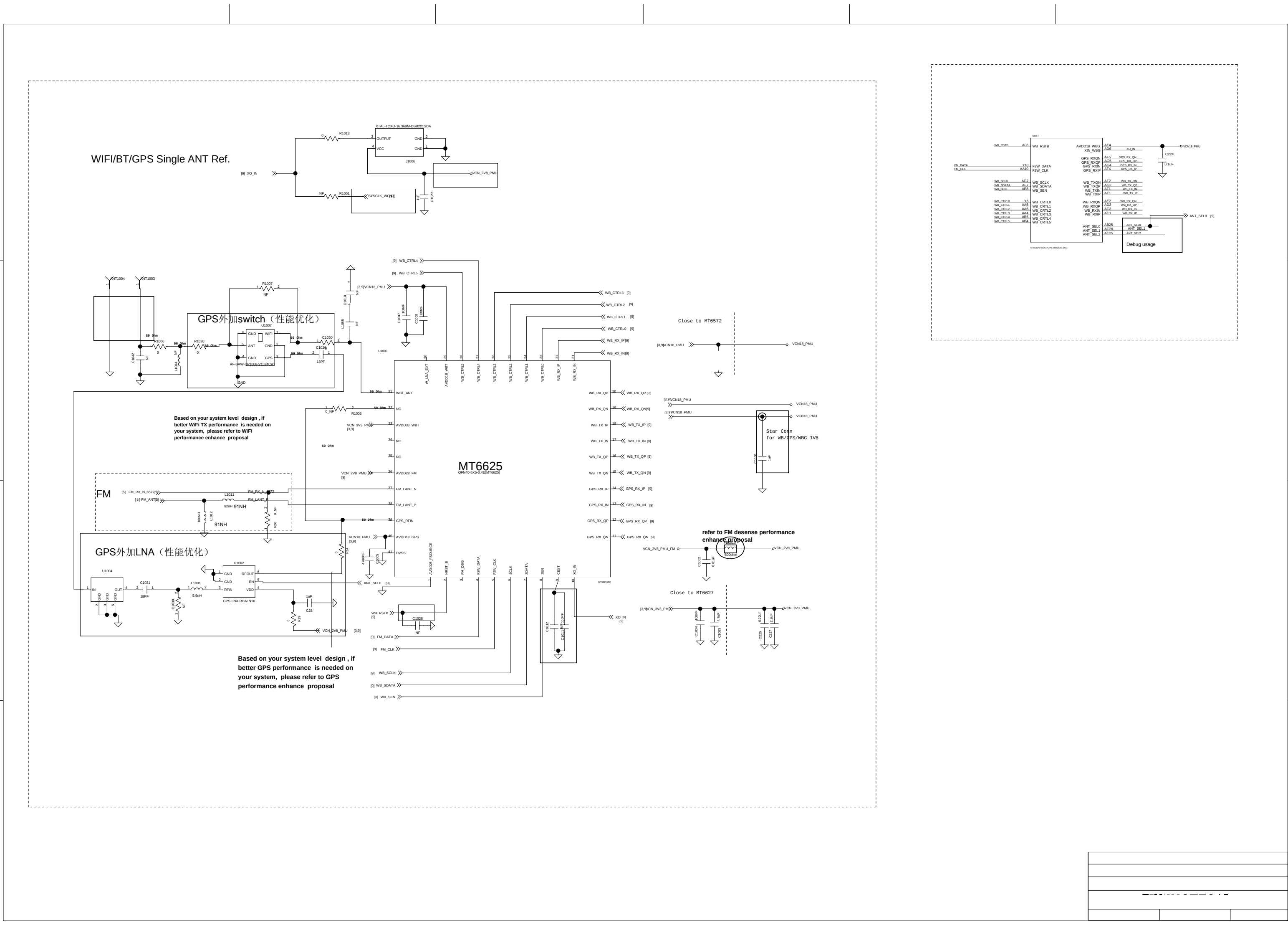
Route AUXADC_REF/AUXADC_TSX as differential trace with well GND shielding and route AUXADC_GND with 24mil trace width under AUXADC_TSX/AUXADC_REF trace to provide return current path.

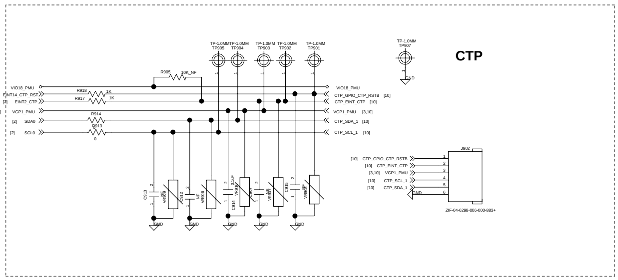
MODE \ Logic	DCXO_32K_EN	XMODE	VXODIG
DCXO + 32K XO	0(GND)	1(VIO18)	1(VIO18)
DCXO + 32K-Less	1(VTXCO28)	1(VTXCO28)	1(VTXCO28)

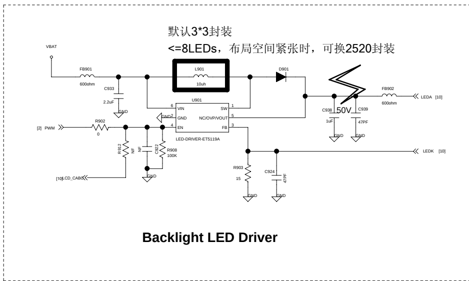
DOCUMENT NO.: 07_RF_MT6166_TXVR_2G_B34_B39	SIZED: A1
DEPARTMENT: Hardware DEPT.	
COMPANY: WINGTECH	
DESIGNER: <DESIGNER> Last Saved Date: <undefined>	SHEET: 7 OF 17

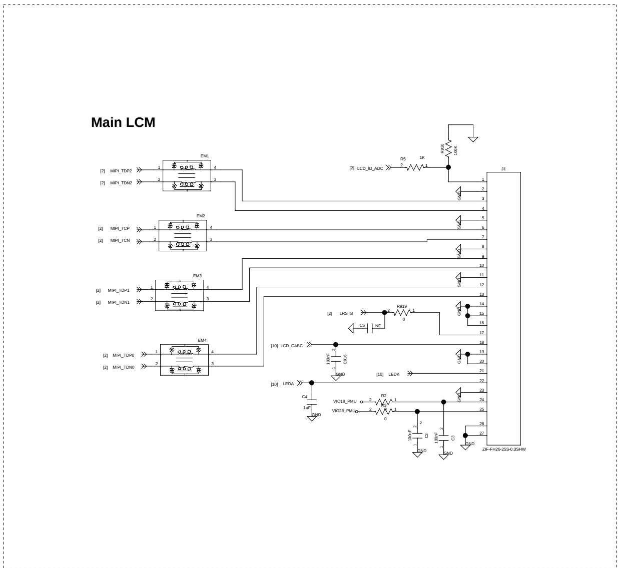


SKY77590 兼容SKY77584设计
SKY77590 pin22 pin24可接其他TRX端口，空载时需要51欧姆接地
SKY77584 pin22 pin24接地
多余的端口，用50欧姆接地
当没有BAND2时，R4=51
SKY77590 control logic table
VctA VctB VctC TxEn
BS2 BS1 mode ebable
(TRX1)G_DCS L L H L
(TRX2)W_Band8 L H H L
(TRX3)W_Band1 H L H L
(TRX4)W_Band5 H H H L
(TRX5)W_Band2 L H L L
(TRX6) NA H L L L
EDGETX_L H L H H
EDGETX_H H H H H
GMSKTX_L H L L H
GMSKTX_H H H L H
EDGE TXM
C708
39pF
J601
IN OUT
RF-SWITCH-818000291
GND
GND
I/O
GND
J602
BPI0~6 are 2G+3G mode both
BPI7~13 are 3G mode only
增加R756 0欧姆的兼容，CMOS PA 可用，去掉试项目情况


WIFI/BT/GPS Single ANT Ref.
GPS外加switch（性能优化）
MT6625
Based on your system level design , if better WiFi TX performance is needed on your system, please refer to WiFi performance enhance proposal
FM
GPS外加LNA（性能优化）
Based on your system level design , if better GPS performance is needed on your system, please refer to GPS performance enhance proposal
Close to MT6572
Star Conn for WB/GPS/WBG 1V8
refer to FM desense performance enhance proposal
Close to MT6627
Debug usage

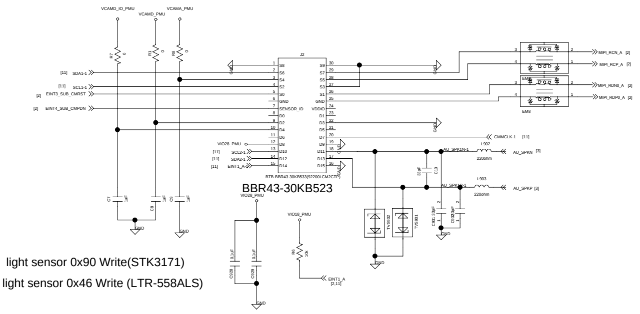

CTP
ZIF-04-6298-006-000-883+


默认3*3封装
<=8LEDs，布局空间紧张时，可换2520封装
50V
Backlight LED Driver


Main LCM

Sub Camera

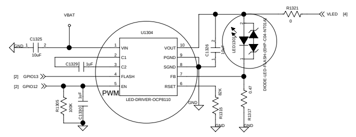
FRONT_camera+LP Sensor+SPK



light sensor 0x90 Write(STK3171)

light sensor 0x46 Write (LTR-558ALS)

```
Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible
```



OCP8111 R1315=68K R1317=0.22

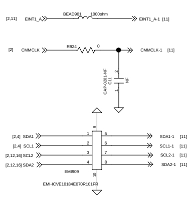
OCP8110 R1315=82K R1317=0.47

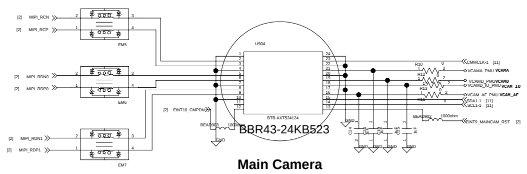
FLASH LED

light sensor 0x90 Write(STK3171)

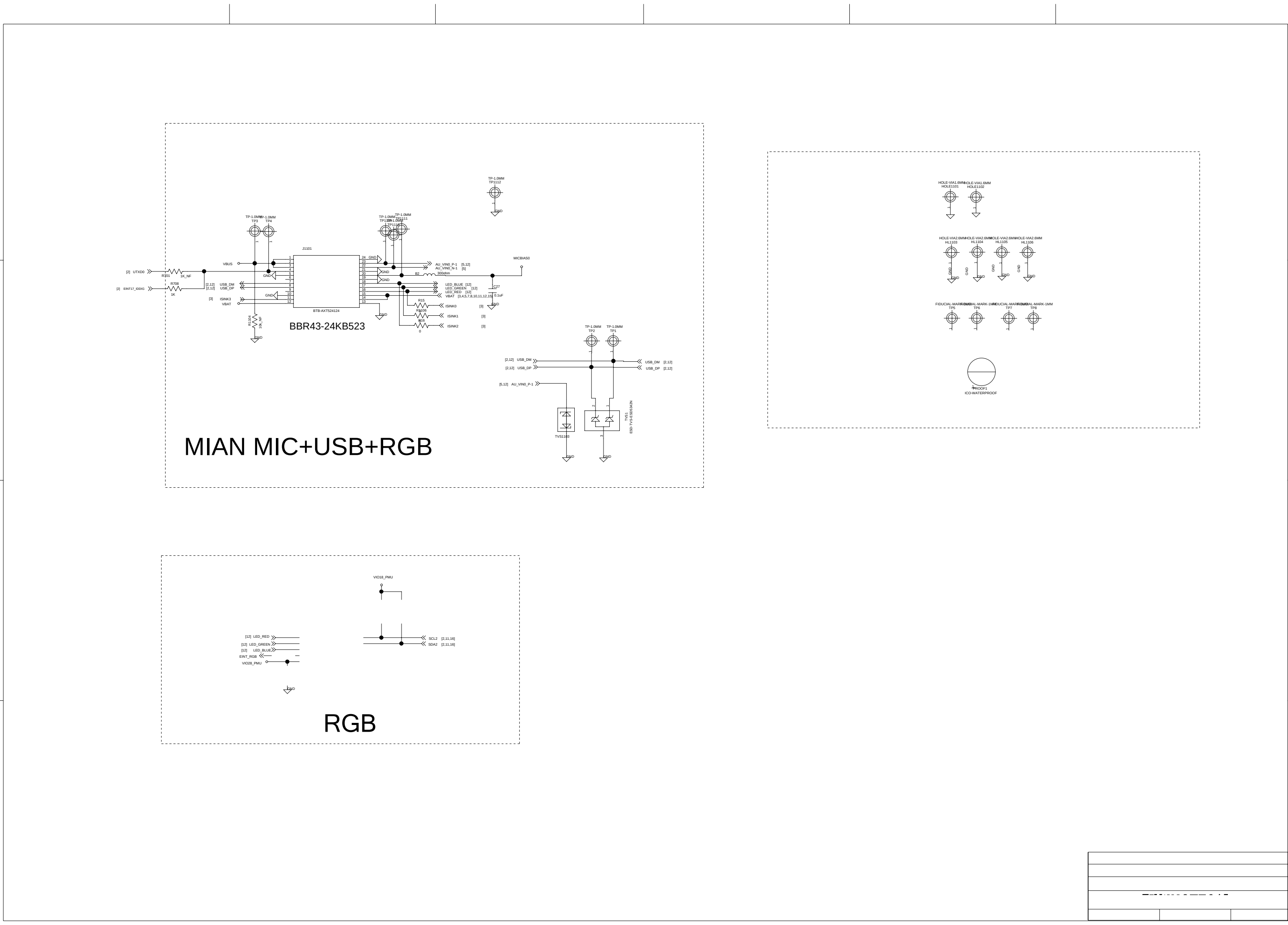
light sensor 0x46 Write (LTR-558ALS)


```
Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible
```

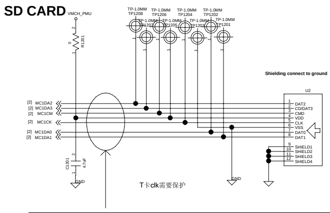





Main Camera

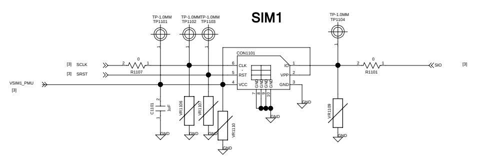

MIAN MIC+USB+RGB
BBR43-24KB523
RGB

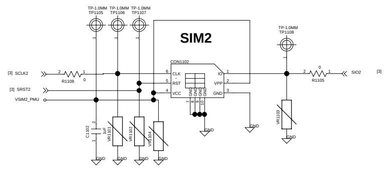
SD CARD

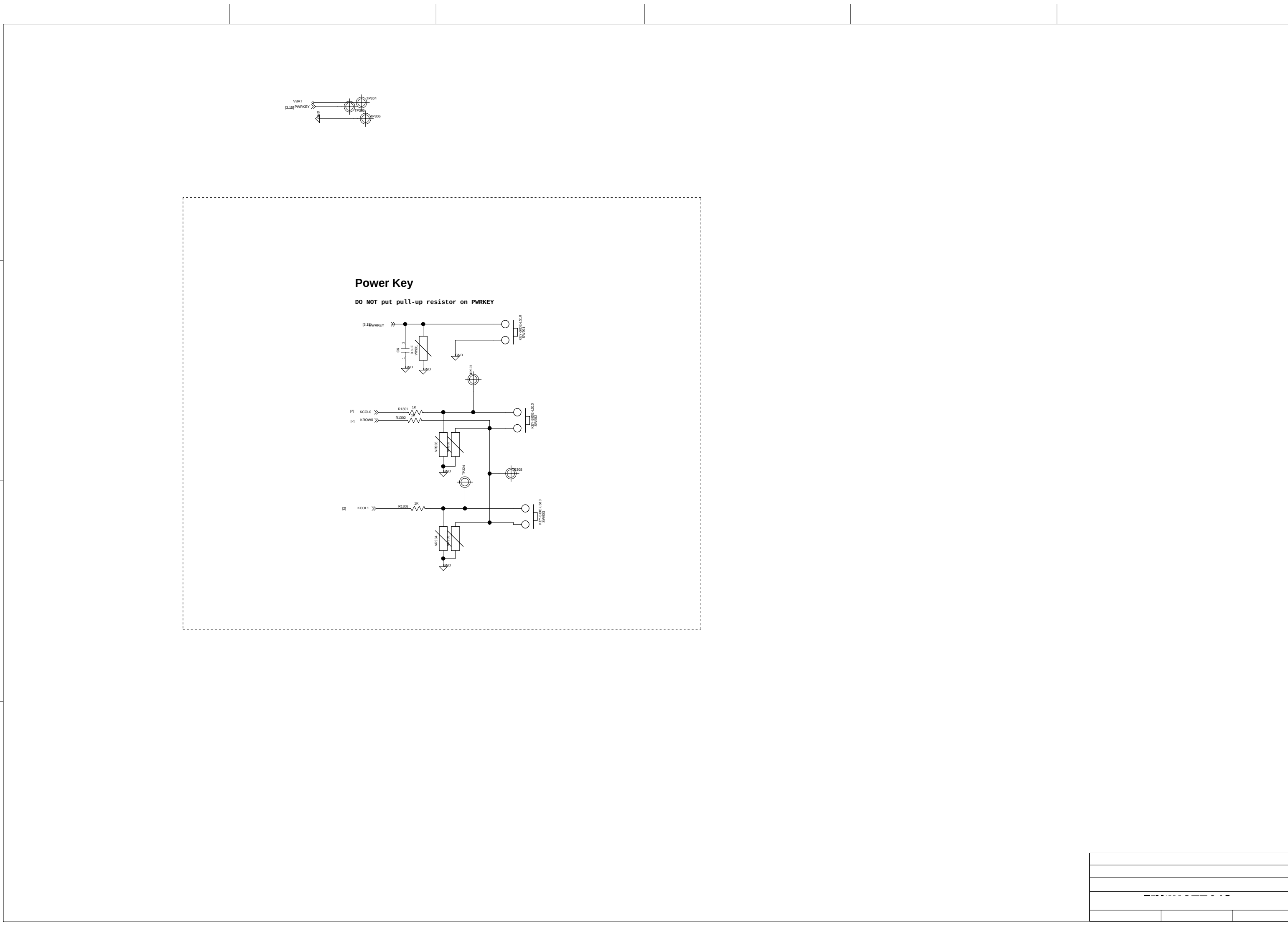


T-card ESD protection is optional depends on
T-card's type and position

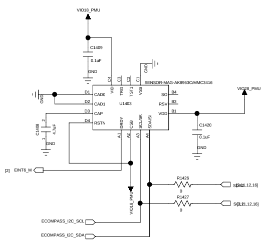
Shielding connect to ground


SIM1


SIM2


Power Key
DO NOT put pull-up resistor on PWRKEY
VBAT
PWRKEY
KCOL0
KROW0
KCOL1

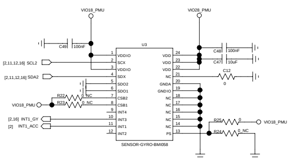
M-Sensor



0x18 Write

Gsensor供货不定尽量双lay,目前商务上建议优选BMA222E

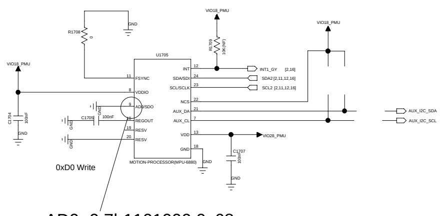
Gyroscope



LSM330 C12=0.01uF(25V)

LSM330 R22 R23=0

Gyroscope



0xD0 Write

AD0=0 7b1101000 0x68

AD0=1 7b1101001 0x69

TITLE:	
DOCUMENT NO.:	
DEPARTMENT:	
COMPANY:	

Version History

Date	Version	Page	Description
2013/03/11	V0.1		Draft release
2013/03/22	V0.11	01	1. Add R107/R103 2. Delete R605/R107
		03	1. Add R326/R328
		04	1. Delete U1802 and related circuit 2. Add U401 and related circuit
		06	1.Add R418
		07	1.Delete R639/R634
		10	1. Swap U2 (MT6627) pin 36, 37 and 38 and their circuitry. 2. Remove R1070, R1073 3. F201 change footprint to DIPLEXER/6P/SMD/DP1608
		15	1. Change C1218,C1219 to 22pF, Change C1214,C1215 to 1.2pF, Change C1216,C1217 to 68pF , Change c1203 4.7uF
		1	1. Remove C138, c153, C145. Change C140, C148 to 1u
		3	1. Add C353, R339, C243 2. Modify Isense trace.
		7 8 9	1. Update RF matching (L615, L600, L619, L616, L622, C670, L623, L626, L625, C629, C627, C603, C662, C664, L617, C624, R617, C651, C655, C652, L610, C648, C657, L609, L608, L611, C649, C626, L607, R620, C634, L631, L627, L629, L636)
2013/03/22	V0.12	1 3 6 12	1. Add R360~R367, R101, R108, R105, R106, R109, R102, R104, R116, R117, R118, R115, R114, R113, R111 R418, R512, R513, R915, R916 for low-power testing
		10	1. C222 change to 4.7uF
		12	1. Add C914 / R904